Robust API Error Handling

Mini Assessment · Defensive JSON and Retry Design

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Instructions

Answer in your own words. Use short examples or pseudocode where helpful. Explain the reasoning behind each decision, not just the name of a technique.

Question 1 · Rate limiting (HTTP 429)

A service returns HTTP 429 during a batch request. Describe a safe retry strategy. Address Retry-After, exponential backoff, retry limits, and what the program should do when retries are exhausted.

Question 2 · Invalid or unexpected response body

A request succeeds at the transport layer, but parsing the response as JSON raises an exception. Describe how you would inspect and handle the response without crashing the whole application. Include what should be logged and what should be shown to the caller.

Question 3 · Missing fields in JSON

Given a JSON object where an optional key may be absent, compare direct indexing with a safe lookup that provides a default. Give a short example and explain when a missing value should be treated as an error.

Question 4 · Combine the safeguards

Write pseudocode for a function that requests JSON from an API, handles a rate limit, validates the response, and returns a safe result or a clear error. Show where retry limits and parsing checks belong.

Question 5 · Test plan

List at least four test cases for your function. Cover a successful response, a rate limit, malformed JSON, and a missing required or optional field. For each case, state the expected behavior.

Question 6 · Design reflection

Describe one risk of retrying too aggressively and one way to make API error handling easier to debug in production.

Self-check

Before submitting, confirm that your answers distinguish retryable from non-retryable failures, avoid exposing secrets in logs, and define what the caller receives on failure.

Student confirmation: I reviewed and completed these answers myself.